Hadoop Project (HW #2)

1. Set up a single-node Hadoop cluster according to the Hadoop reference manual provided.

You may use the WordCount.java source file exactly as included in the attached file.

Note: If you are not familiar with installing Linux and are worried about the process, it is recommended to download a free VM program such as Oracle VirtualBox, install it on Windows, and then install Linux on top of it. This is the safest method.

2. The actual dataset for the assignment (2006.csv) has been uploaded to the course board. Because of the 100MB attachment limit, the compressed tar.gz file was split into two parts (2006.tar.gzaa, 2006.tar.gzab). Combine the two files and then extract the archive before using the dataset. After decompression, the dataset will be approximately 672MB.

- E.g.,) Combine split archive files: cat 2006.tar.gza* > 2006.tar.gz
- E.g.,) Extract the combined archive: tar -xzvf 2006.tar.gz

3. Run WordCount in Hadoop.

When running the job, make sure to redirect (2>) the progress output displayed on the terminal into the assignment output file.

E.g.,): hadoop jar ./wordcount.jar WordCount /user/dpark/wordcount/input /user/dpark/wordcount/output **2>&1 | tee** output_YourName_StudentID.txt

4. After Hadoop finishes running, append (>>) the final wordcount output data to the same assignment output file created above.

E.g.,): hadoop fs -cat /user/dpark/wordcount/output/* **>>** output_YourName_StudentID.txt

This will create a single file containing both the progress logs and the final output results.

5. When you open the output file generated on Linux in Windows, the line-break format will be broken. Therefore, convert the file as shown below before submitting (-5 points if not converted).

E.g.,): todos output_YourName_StudentID.txt

If the tofrodos package is not installed on Ubuntu, install it first and then run the conversion command.

E.g.,): sudo apt install tofrodos

[Submission Items]

1. The **output file** generated by redirecting the execution results (Filename: output_YourName_StudentID.txt)

2. When submitting online, include the following two explanations directly in the submission text field (no separate file required):
   A. After setting up Hadoop according to the manual, running WordCount on a small dataset (few KB ~ few MB), as shown in the example, should work without issues. However, when running a large dataset such as 2006.csv, Hadoop may fail due to system limitations. If an error occurs, briefly describe the error in e-class. (If the job runs successfully without issues, state "No issues." In this case, you must provide evidence through your output file). It is common that small datasets run correctly, but large datasets like 2006.csv may generate errors. If an error occurs, describe what caused it, how you fixed it, and how you successfully ran the job with 2006.csv. This is the main goal of the assignment.
   B. If an error occurred while processing 2006.csv, describe how you resolved it in your online submission.

[Deadline]: Due by November 19 (Thu) before class (15:00). Late submissions will receive a score of 0.

Online submission only.